Shorted–Gram Reassembly of an Incomplete High–Prime Image at a Fixed Shift: Missing Walsh Coordinates, Exact Kernel Tests, and Endpoint Certificates

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Abstract

The critical-cutoff analysis of TPC–59 separates the complete high-prime Walsh face into coordinates represented by the terminal census and high coordinates absent from that image. Restoring the latter is not a monotone operation: their native output may cancel the already represented affine vector. We formulate this reassembly problem on the literal finite row-coefficient space at one fixed nonzero shift.

For linear maps $A, M : \mathcal{K} \to \mathcal{H}$, representing respectively the represented affine output and the missing-high correction, put $B = A + M$ and

$$\delta_{\mathrm{reasm}} = \inf_{Az\neq 0} \frac{\|Bz\|^2}{\|Az\|^2}.$$

The denominator does not see $N = \operatorname{Ker} A$, while $B(N) = M(N)$ can cancel the full output. We prove that the exact quotient is obtained only after shorting B^*B against these nuisance directions. If $S = N^\perp$, $B_1 = B|_S$, $B_0 = B|_N$, and P projects onto $B(N)^\perp$, then

$$\delta_{\mathrm{reasm}} = \inf_{s\in S\setminus\{0\}} \frac{\|PB_1 s\|^2}{\|As\|^2}.$$

Equivalently, it is the least generalized eigenvalue of the pair $(Q, A_S^* A_S)$, where the shorted numerator is

$$Q = B_1^* B_1 - B_1^* B_0 (B_0^* B_0)^\dagger B_0^* B_1.$$

This yields the exact finite-dimensional positivity test $\operatorname{Ker}(A+M) \subseteq \operatorname{Ker} A$, an anti-diagonal image test, and a rank kill criterion. A naive compression to $N^\perp$ is shown by sharp examples to overestimate the transfer.

We also give a distinguished-vector signed-defect identity and a sharp raw-mass certificate. If each native output receives at most d_{miss} missing atoms, then

$$\|Mz\|^2 \le d_{\mathrm{miss}} \mathcal{D}_{\mathrm{miss}}(z).$$

Consequently a missing-mass saving $X^{-\sigma_{\mathrm{miss}}}$ automatically defeats a fiber-degree cost $X^{\mu_{\mathrm{miss}}}$ under the strict exponent inequality $\sigma_{\mathrm{miss}} > \mu_{\mathrm{miss}}$, relative to the represented output. At exponent equality, finer constant or subpower information decides whether this certificate still has zero fixed-power cost. The terminal image and its complement are identified on the same physical Walsh carrier and raw normalization. No positive bound for the resulting arithmetic constant is proved. Thus the work closes the L0 linear-algebra interface and the L1 fixed-shift dictionary, but proves no parity improvement, prime-pair asymptotic, or twin-prime consequence.

Keywords: shorted operator; generalized Schur complement; missing Walsh coordinates; fixed physical shift; fiber degree; parity boundary.

Convention. One nonzero physical shift h_0, one direct residue/orbit slice, and one side of the inherited Walsh system are fixed unless an explicit direct sum is displayed. Atomic Hilbert spaces use raw counting measure. Baseline carriers are fixed independently of the bounded row coefficient vector, and only coordinates identically zero on that coefficient class are removed.

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1 Introduction

The large-prime branch of the TPC program begins with an exact Walsh expansion, resolves a unique very-high prime, transports the resulting atoms through residual fibers, and finally recombines them into the native affine output at a fixed physical shift $h_0 \neq 0$ [4–7]. TPC–59 makes a necessary correction to that chain. Its largest-prime census injects the retained terminal atoms into the high Walsh carrier, but it need not cover the full carrier. With the notation of that paper,

$$\mathcal{D}_0 = \mathcal{D}_{\mathrm{base}} + \mathcal{D}_{\mathrm{miss}} + \mathcal{D}_H^{\mathrm{rep}}. \tag{1}$$

Here $\mathcal{D}_H^{\mathrm{rep}}$ is the represented high diagonal, while $\mathcal{D}_{\mathrm{miss}}$ is high Walsh mass outside the terminal image [8].

This distinction survives the final grouping. Let g_{rep} be the native affine vector built from the base column and the represented high column, and let h_{miss} be the output of the missing high coordinates. The full physical vector is

$$g_{\mathrm{full}} = g_{\mathrm{rep}} + h_{\mathrm{miss}}. \tag{2}$$

There is no inequality $\|g_{\mathrm{full}}\| \geq \|g_{\mathrm{rep}}\|$: the omitted vector can be destructively aligned. Nor does small raw missing mass by itself solve the problem, because many small missing atoms may enter one native output with the same phase.

The purpose of this paper is to identify exactly what must be proved at this reassembly arrow. The answer contains a subtle quotient. Two row coefficient vectors may have the same represented output while having different missing output. Therefore the full map need not descend to the quotient by the kernel of the represented map. The correct uniform constant minimizes over that entire kernel fiber. In matrix language this is a shorted operator, or equivalently a generalized Schur complement [1, 3].

1.1 The three questions

Fix finite-dimensional Hilbert spaces $\mathcal{K}$ and $\mathcal{H}$, and linear maps

$$A, M : \mathcal{K} \longrightarrow \mathcal{H}, \qquad B := A + M. \tag{3}$$

The TPC specialization is:

- A is the base plus represented-high affine map;
- M is the missing-high native grouping;
- B is the full declared affine output on the same fixed slice.

We answer three separate questions.

1. What is the optimal coefficient-uniform ratio

$$\delta_{\mathrm{reasm}} := \inf_{Az \neq 0} \frac{\|Bz\|^2}{\|Az\|^2}? \tag{4}$$

2. What exact kernel, image, or rank condition kills that uniform gate?
3. What arithmetic estimate on missing raw mass and native fiber degree is sufficient for a distinguished-vector or uniform lower transfer?

These questions must not be conflated. A distinguished physical vector can reassemble safely even when the uniform constant vanishes. Conversely, the qualitative kernel condition can hold while the least ratio tends to zero with X. Finally, a raw atomic estimate does not determine the sign of the represented–missing cross term.

1.2 Main results

Let

$$N := \operatorname{Ker} A, \qquad S := N^{\perp}, \qquad W := B(N) = M(N), \tag{5}$$

and let $P_W^{\perp}$ denote the orthogonal projection onto $W^{\perp}$. The first result is the exact nuisance-elimination formula

$$\boxed{\delta_{\text{reasm}} = \inf_{s \in S \setminus \{0\}} \frac{\|P_W^{\perp} Bs\|^2}{\|As\|^2}.} \tag{6}$$

Thus the effective map is not $B|_S$, but its projection away from the output that can be generated freely by N. We derive the equivalent shorted-Gram and least-singular-value formulas and prove

$$\boxed{\delta_{\text{reasm}} > 0 \quad \Longleftrightarrow \quad \operatorname{Ker} B \subseteq \operatorname{Ker} A.} \tag{7}$$

The equivalence is finite-dimensional. It is an exact L0 statement; it does not assert the condition for the TPC coefficient family.

For the literal missing atomic lift, let $\mathcal{F}_i^{\text{miss}}$ be the missing atoms entering native output i, and put

$$d_{\text{miss}} := \max_i |\mathcal{F}_i^{\text{miss}}|, \qquad \mathcal{D}_{\text{miss}}(z) := \sum_i \sum_{u \in \mathcal{F}_i^{\text{miss}}} |m_u(z)|^2. \tag{8}$$

Then the grouping operator has squared norm d_{miss}, and hence

$$\|Mz\|^2 \le d_{\text{miss}} \mathcal{D}_{\text{miss}}(z). \tag{9}$$

Together with the reverse triangle inequality this gives the sharp certificate

$$\frac{\|Bz\|^2}{\|Az\|^2} \ge \left(1 - \sqrt{\frac{d_{\text{miss}} \mathcal{D}_{\text{miss}}(z)}{\|Az\|^2}}\right)_+^2. \tag{10}$$

We exhibit a family saturating the threshold: $d_{\text{miss}} = n$, raw missing mass $1/n$, and exact cancellation of a unit represented output.

1.3 Claim boundary and route position

The paper deliberately stops at the first honest boundary. The shorted-Gram theorem, signed-defect identity, and mass–degree estimate are L0. Their construction from the complete high Walsh carrier, the represented terminal image, and its complement at one fixed h_0 is L1. No estimate here proves that the missing-mass saving exceeds the fiber-degree cost, that the effective map is injective uniformly in X, or that the distinguished physical vector avoids cancellation. Those are new arithmetic gates.

Accordingly, no statement in this paper is an L2 parity advance. In particular, we prove neither a positive fixed-h_0 prime-pair lower bound nor an asymptotic for twin primes. The value of the result is diagnostic: it replaces an informal “restore the omitted coordinates” arrow by an optimal constant, an exact zero test, and two quantitatively auditable routes to positivity.

2 The represented, missing, and full physical maps

We first place all three outputs on one coefficient space. This prevents two common changes of problem: choosing the missing vector independently of the represented vector, and comparing norms formed from different atomic normalizations.

2.1 Fixed carriers and the literal high partition

Let $\mathcal{W}_0$ be the complete declared Walsh slice of TPC–59, after removing only coordinates which are identically inactive throughout the fixed bounded coefficient class. Throughout, y is TPC–59 very-high admissible:

$$y > \max\{|h_0|, D_+, J_+, \Delta_M, \sqrt{N_+}\}, \qquad \frac{D_+N_+}{y} < D_-N_-, \tag{11}$$

and every retained source row has its inherited unique representation $m = d_m\ell_m$ in the declared one-side row system [8]. Let $\mathcal{W}_H(y)$ be the resulting very-high part. The fixed-shift terminal census supplies an injection

$$\iota : \mathcal{U}^{\mathrm{term}}_{h_0,>y} \longrightarrow \mathcal{W}_H(y). \tag{12}$$

Set

$$\mathcal{W}_B := \mathcal{W}_0 \setminus \mathcal{W}_H(y), \qquad \mathcal{W}_R := \operatorname{Ran}\iota, \qquad \mathcal{W}_M := \mathcal{W}_H(y) \setminus \mathcal{W}_R. \tag{13}$$

The letters B, R, M stand for base, represented high, and missing high. They form the disjoint partition

$$\mathcal{W}_0 = \mathcal{W}_B \,\dot\cup\, \mathcal{W}_R \,\dot\cup\, \mathcal{W}_M. \tag{14}$$

Let $\mathcal{K}_X$ be the raw finite row-coefficient Hilbert space. For each Walsh atom w, literal factorization gives

$$b_w(\zeta) = \beta_w\zeta_{m(w)}, \qquad \zeta \in \mathcal{K}_X, \tag{15}$$

with β_w fixed by the declared row, orbit, cutoff, and profile data. No favorable atom-dependent coefficient is introduced. The raw atomic diagonals are

$$\mathcal{D}_E(\zeta) := \sum_{w\in\mathcal{W}_E} |b_w(\zeta)|^2, \qquad E \in \{B, R, M\}. \tag{16}$$

In the abstract atomic notation used below, the missing lift is normalized by the same physical amplitudes, so

$$\mathcal{D}_{\mathrm{miss}}(\zeta) = \mathcal{D}_M(\zeta). \tag{17}$$

Therefore

$$\boxed{\mathcal{D}_0 = \mathcal{D}_B + \mathcal{D}_R + \mathcal{D}_M,} \qquad \mathcal{D}_R = \sum_k \mathcal{D}_k \tag{18}$$

on the critical high face, with the dyadic cells of TPC–59. This is a Pythagorean identity in the atomic space; it is not an additivity statement for the coherently grouped output.

2.2 Recovery of the physical labels

For a nonzero high Walsh atom w, the literal fixed-shift dictionary recovers a unique triple (m, j, γ), the target

$$n = mj + h_0, \tag{19}$$

and, because $y > \sqrt{N_+}$, the unique factorization

$$p = P^+(n) > y, \qquad r = \frac{n}{p}, \qquad s = d_m r. \tag{20}$$

The native output key is denoted $i = i(w)$; on the displayed left slice it includes the side, orbit tag, and row-time data, and the right slice is handled by orthogonal direct sum. We write

$$q(w) = (i(w), s(w)). \tag{21}$$

In particular, the native key i fixes the side, the opposite row, and the orbit time j. All of these labels are recovered from the physical atom. They are not new choices made after seeing ζ.

For $E \in \{R, M\}$, define the sign-compatible high coefficient by

$$z_E(i, s) := -\sum_{\substack{w \in \mathcal{W}_E \\ q(w)=(i,s)}} b_w. \tag{22}$$

Let

$$(C_\mu z)_i := \sum_s \mu(s) z(i, s), \qquad h_R := C_\mu z_R, \qquad h_M := C_\mu z_M. \tag{23}$$

The minus sign in (22) is the fixed high-character sign; the remaining $\mu(s)$ is the low-label gauge. Thus these formulas use the TPC–58 sign convention and do not apply a second Möbius sign.

For the base carrier write

$$(v_0)_i := \sum_{\substack{w \in \mathcal{W}_B \\ i(w)=i}} \mu(u_w) b_w, \tag{24}$$

where u_w is its low Walsh label. The precise inherited support masks are already contained in $\mathcal{W}_B$.

Proposition 2.1 (Literal three-map identity)**.** *There are linear maps*

$$A_X, M_X, B_X : \mathcal{K}_X \longrightarrow \mathcal{H}_{\text{out}} \tag{25}$$

such that, for every physical row coefficient vector ζ,

$$A_X\zeta = v_0(\zeta) + h_R(\zeta), \tag{26}$$
$$M_X\zeta = h_M(\zeta), \tag{27}$$
$$B_X\zeta = v_0(\zeta) + h_R(\zeta) + h_M(\zeta) = (A_X + M_X)\zeta. \tag{28}$$

Moreover $h_R = C_{\mu,H}T_H^{\text{rep}}\zeta$ is exactly the TPC–58 high map restricted to $\mathcal{W}_R = \operatorname{Ran}\iota$, while h_M is the same physical grouping restricted to $\mathcal{W}_M$. Consequently

$$\mathcal{N}_R(\zeta) := \|A_X\zeta\|_2^2, \qquad \mathcal{N}_{\text{full}}(\zeta) := \|B_X\zeta\|_2^2 \tag{29}$$

are the represented and full native affine energies on one and the same declared slice.

Proof. Linearity follows from (15) and the fixed carrier sums. The gauge alignment of TPC–58 identifies the represented terminal output with $C_{\mu,H}T_H^{\text{rep}}\zeta$. Restricting the identical atom-to-native map to the complementary high set defines M_X. The disjoint coordinate partition (14) then gives the last identity before any output norm is taken. □

Remark 2.2 (Why the atom partition does not imply output monotonicity)*.* The equality $\mathcal{D}_0 = \mathcal{D}_B + \mathcal{D}_R + \mathcal{D}_M$ is taken before native grouping. After grouping,

$$\mathcal{N}_{\text{full}} - \mathcal{N}_R = \|h_M\|^2 + 2\operatorname{Re}\langle A_X\zeta, h_M\rangle.$$

The cross term has no fixed sign. Thus restoring a disjoint atomic coordinate set can decrease the physical output norm.

2.3 Prime-resolved alias structure

To locate the possible cancellation more precisely, decompose

$$h_R = -\sum_{p>y} v_p^R, \qquad h_M = -\sum_{p>y} v_p^M, \tag{30}$$

where, for $E \in \{R, M\}$,

$$(v_p^E)_i := \sum_{\substack{w\in\mathcal{W}_E\\ i(w)=i,\ p(w)=p}} \mu(s(w))b_w. \tag{31}$$

On the active very-high carrier, squarefree support gives $|\mu(s(w))| = 1$. Hence no modulus is lost when the raw atom is placed in a prime-resolved component.

Proposition 2.3 (Same-prime separation and unequal-prime alias)**.** *For every fixed pair (i, p), there is at most one active high Walsh atom. Consequently*

$$\langle v_p^R, v_p^M\rangle = 0 \tag{32}$$

for every p, and

$$\sum_p (\|v_p^R\|^2 + \|v_p^M\|^2) = \mathcal{D}_R + \mathcal{D}_M. \tag{33}$$

The exact reassembly defect is

$$\begin{aligned}\mathcal{N}_{\text{full}} - \mathcal{N}_R = {} & \left\|\sum_q v_q^M\right\|^2 - 2\,\mathrm{Re}\sum_q \langle v_0, v_q^M\rangle \\ & + 2\,\mathrm{Re}\sum_{\substack{p,q>y\\ p\neq q}} \langle v_p^R, v_q^M\rangle.\end{aligned} \tag{34}$$

Thus represented–missing cancellation cannot occur at the same prime; the remaining dangers are base–missing interaction and unequal-prime alias at a common native output.

Proof. Fix (i, p). Since i fixes j, suppose that p meets two source rows m, m'. Then $p \mid (m-m')j$. The alternative $p \mid j$ is excluded by $p > J_+$; hence $p \mid m - m'$. But $|m - m'| \le \Delta_M < p$, so $m = m'$, and the inherited unique row representation recovers the same full source atom. This is exactly the TPC–59 critical-cutoff fixed-(j, p) argument [8]. Thus a fixed (i, p) contains at most one high atom. Such an atom belongs to exactly one of $\mathcal{W}_R, \mathcal{W}_M$, proving disjoint coordinate support and (32). Summing the coordinate squares gives (33). Finally expand

$$\|v_0 - \sum_p v_p^R - \sum_q v_q^M\|^2 - \|v_0 - \sum_p v_p^R\|^2$$

and delete the same-prime cross terms using (32). □

The proposition is useful but does not close reassembly. Different high primes can land at the same native output, and degree two already permits exact cancellation. Prime resolution is therefore a diagnostic layer, not a substitute for the full native Gram.

2.4 The six-item physical audit

The construction above passes the following interface audit.

1. The coefficients are the literal $b_w = \beta_w \zeta_{m(w)}$.
2. One fixed physical h_0 appears in $n = mj + h_0$.

3. Every atomic norm is the raw counting norm; no fiber average is inserted.
4. The atom representation is nonduplicated, and the row space removes only identically null columns. A quotient by $\operatorname{Ker} A_X$ is not taken unless the full map actually descends to it.
5. The active supports are exactly $\mathcal{W}_B, \mathcal{W}_R, \mathcal{W}_M$ inside the declared Walsh carrier; coefficient-dependent zeros do not redefine them.
6. Any eventual reassembly loss remains an explicit term in the strict endpoint budget $< 1/400$.

3 Exact nuisance elimination and the shorted Gram

This section is finite-dimensional linear algebra. It applies at every fixed scale X, after which one may ask whether the resulting constants are uniform or polynomially controlled as $X \to \infty$.

Let $\mathcal{K}, \mathcal{H}$ be finite-dimensional complex Hilbert spaces and let

$$A, M : \mathcal{K} \longrightarrow \mathcal{H}, \qquad B := A + M. \tag{35}$$

Assume $A \neq 0$, so that the quotient below has a nonempty domain, and define

$$\delta(A, M) := \inf_{Az \neq 0} \frac{\|Bz\|^2}{\|Az\|^2}. \tag{36}$$

3.1 The invisible directions

Put

$$N := \operatorname{Ker} A, \qquad S := N^{\perp}, \qquad A_S := A|_S, \tag{37}$$

and let

$$B_S := B|_S, \qquad B_N := B|_N = M|_N, \qquad W := \operatorname{Ran} B_N. \tag{38}$$

Write

$$P := P_{W^{\perp}} = I - B_N B_N^{\dagger}. \tag{39}$$

Here † denotes the Moore–Penrose inverse, and $B_N B_N^{\dagger}$ is the orthogonal projection onto W [1].

Every $z \in \mathcal{K}$ decomposes uniquely as $z = s + k$, with $s \in S$ and $k \in N$. The denominator of (36) depends only on s, whereas the numerator is

$$\|B_S s + B_N k\|^2. \tag{40}$$

Thus k is a nuisance direction which can be chosen to cancel the output without changing the represented norm.

Theorem 3.1 (Exact nuisance elimination)**.** *For the maps in* (35)*,*

$$\boxed{\delta(A, M) = \inf_{s \in S \setminus \{0\}} \frac{\|P B_S s\|^2}{\|A_S s\|^2}.} \tag{41}$$

For fixed s, all minimizing nuisance vectors are

$$k_s = -B_N^{\dagger} B_S s + k_0, \qquad k_0 \in \operatorname{Ker} B_N. \tag{42}$$

Proof. For fixed s, minimizing (40) over $k \in N$ is the least-squares problem of approximating $-B_S s$ by $\operatorname{Ran} B_N$. Hence

$$\begin{aligned} \inf_{k \in N} \|B_S s + B_N k\|^2 &= \operatorname{dist}(B_S s, W)^2 \\ &= \|P B_S s\|^2. \end{aligned} \tag{43}$$

The Moore–Penrose least-squares formula gives (42). Since A_S is injective, $A_S s \neq 0$ precisely when $s \neq 0$. Taking the infimum over s proves (41). $\square$

3.2 Schur complement and shorted operator

Let $G := B^*B$, and write its block matrix relative to $\mathcal{K} = S \oplus N$:

$$G = \begin{pmatrix} G_{SS} & G_{SN} \\ G_{NS} & G_{NN} \end{pmatrix} = \begin{pmatrix} B_S^*B_S & B_S^*B_N \\ B_N^*B_S & B_N^*B_N \end{pmatrix}. \tag{44}$$

Because $G \geq 0$, its blocks satisfy the standard range compatibility needed for the generalized Schur complement. Define

$$Q := G_{SS} - G_{SN}G_{NN}^{\dagger}G_{NS}. \tag{45}$$

Proposition 3.2 (Shorted-Gram identity)**.** *One has*

$$\boxed{Q = B_S^*PB_S \geq 0.} \tag{46}$$

Moreover the short of G to S is

$$\mathrm{short}_S(G) = \begin{pmatrix} Q & 0 \\ 0 & 0 \end{pmatrix}. \tag{47}$$

Proof. The Moore–Penrose projection identity gives

$$B_N(B_N^*B_N)^{\dagger}B_N^* = B_NB_N^{\dagger} = P_W. \tag{48}$$

Substitution into (45) yields

$$Q = B_S^*(I - P_W)B_S = B_S^*PB_S.$$

The block formula for the finite-dimensional shorted positive operator is then exactly (47); see W. N. Anderson and Trapp [3]. □

Let

$$R := A^*A, \qquad R_S := A_S^*A_S. \tag{49}$$

The operator R_S is positive definite on S.

Theorem 3.3 (Exact shorted generalized eigenvalue)**.** *The optimal reassembly constant is*

$$\boxed{\delta(A, M) = \lambda_{\min}\left(R_S^{-1/2}QR_S^{-1/2}\right).} \tag{50}$$

Equivalently,

$$\delta(A, M) = \lambda_{\min}\left(R^{\dagger/2}\,\mathrm{short}_S(B^*B)R^{\dagger/2}\Big|_S\right). \tag{51}$$

Proof. By Theorem 3.1 and Proposition 3.2,

$$\delta(A, M) = \inf_{s \neq 0} \frac{\langle Qs, s\rangle}{\langle R_Ss, s\rangle}.$$

Set $x = R_S^{1/2}s$ and apply the Rayleigh–Ritz principle [2]. This proves (50). On S, the restriction of $R^{\dagger/2}$ is $R_S^{-1/2}$, while it vanishes on N, giving (51). □

3.3 Effective output map

Let $E := \operatorname{Ran} A$. The map

$$A_S : S \longrightarrow E \tag{52}$$

is a Hilbert-space isomorphism, with inverse $A^\dagger|_E$. Define

$$C_{\mathrm{eff}} := PB_S A_S^{-1} : E \longrightarrow W^\perp. \tag{53}$$

Corollary 3.4 (Effective singular value)**.** *One has*

$$\boxed{\delta(A, M) = \sigma_{\min}(C_{\mathrm{eff}})^2.} \tag{54}$$

Thus reassembly consists of choosing a represented output $y \in E$, taking its canonical coefficient preimage $A_S^{-1}y$, forming the full output, and projecting away everything producible by an A-invisible coefficient direction.

Proof. Substitute $y = A_S s$ in (41). □

3.4 Why ordinary compression is wrong

If one fixes the canonical representative $s \in S$ and forbids motion along N, one instead computes

$$\delta_{\mathrm{can}} := \inf_{s \neq 0} \frac{\|B_S s\|^2}{\|A_S s\|^2} = \lambda_{\min}(R_S^{-1/2} G_{SS} R_S^{-1/2}). \tag{55}$$

Since P is a contraction,

$$0 \le \delta(A, M) \le \delta_{\mathrm{can}}, \tag{56}$$

and the inequality may be strict.

Example 3.5 (Canonical compression can miss total cancellation)**.** Take $\mathcal{K} = \mathbb{C}^2$, $\mathcal{H} = \mathbb{C}$, and

$$A(x, k) = x, \qquad M(x, k) = k, \qquad B(x, k) = x + k. \tag{57}$$

Here $S = \{(x, 0)\}$, so the canonical ratio is identically one. However $N = \{(0, k)\}$ and $B(N) = \mathbb{C}$, hence $P = 0$ and $\delta(A, M) = 0$. Explicitly, choosing $k = -x$ annihilates the full output without changing $Ax = x$.

The compressed Gram equals the shorted Gram precisely when the Schur correction vanishes:

$$G_{SN} G_{NN}^\dagger G_{NS} = 0. \tag{58}$$

Because this operator is positive, the condition is equivalent to

$$\operatorname{Ran} B_S \perp \operatorname{Ran} B_N. \tag{59}$$

This operator identity is sufficient, but not necessary, for equality of the two least generalized eigenvalues: a nonzero correction may act only away from the minimizing eigenspace. Thus (58) must not be advertised as an exact criterion for equality of the scalar minima. A particularly safe case is

$$B|_{\operatorname{Ker} A} = 0. \tag{60}$$

Then the full map genuinely descends to $\mathcal{K}/\operatorname{Ker} A$, $W = \{0\}$, and no shorting is needed. Without (60), replacing a coefficient vector by its minimum-norm representative changes the physical full output and is not an admissible simplification.

4 Kernel, anti-diagonal, and rank kill tests

The shorted formula gives an exact qualitative test at each finite scale. It also shows why qualitative injectivity is weaker than the polynomial lower bound needed by an endpoint argument.

Theorem 4.1 (Exact positivity test)**.** *For finite-dimensional $\mathcal{K}, \mathcal{H}$ and $A \neq 0$, the following are equivalent:*

i $\delta(A, M) > 0$*;*

ii $PB_S : S \to W^\perp$ *is injective;*

iii $\operatorname{Ker}(A + M) \subseteq \operatorname{Ker} A$*;*

iv there is no $z \in \mathcal{K}$ *with* $Az \neq 0$ *and* $(A + M)z = 0$*.*

Proof. By Corollary 3.4, positivity is equivalent to injectivity of PB_S on S. If that map is not injective, there is $s \in S \setminus \{0\}$ with $B_S s \in W = \operatorname{Ran} B_N$. Choose $k \in N$ so that $B_N k = -B_S s$. Then $z = s + k \in \operatorname{Ker} B$, while $Az = A_S s \neq 0$. Conversely, any $z = s + k \in \operatorname{Ker} B \setminus \operatorname{Ker} A$ has $s \neq 0$ and $PB_S s = -PB_N k = 0$. This proves all equivalences. □

Corollary 4.2 (Dimension kill criterion)**.** *If*

$$\dim S > \dim \mathcal{H} - \operatorname{rank} B_N, \tag{61}$$

then $\delta(A, M) = 0$*.*

Proof. The codomain $W^\perp$ of PB_S has dimension $\dim \mathcal{H} - \operatorname{rank} B_N$. Under (61), that map cannot be injective. □

For the TPC specialization this supplies a finite-scale kill test which can be checked exactly from the literal matrices. If it fires, no uniform loss exponent λ_{reasm} exists on the declared coefficient class: the ratio is exactly zero, not merely poorly estimated. If it does not fire, no quantitative asymptotic lower bound follows automatically.

4.1 The pair-image formulation

Define the coupled represented–missing map

$$J : \mathcal{K} \longrightarrow \mathcal{H} \oplus \mathcal{H}, \qquad Jz := (Az, Mz), \tag{62}$$

and its actual feasible pair image

$$\mathcal{F}_{A,M} := \operatorname{Ran} J. \tag{63}$$

The physical sum map is

$$D_+(g, h) := g + h. \tag{64}$$

When $\mathcal{F}_{A,M} \neq \{0\}$, put

$$\delta_{\text{pair}} := \inf_{(g,h) \in \mathcal{F}_{A,M} \setminus \{0\}} \frac{\|g + h\|^2}{\|g\|^2 + \|h\|^2}. \tag{65}$$

Here $\mathcal{F}_{A,M}$ carries the norm inherited from the Hilbert direct sum $\mathcal{H} \oplus \mathcal{H}$.

Proposition 4.3 (Anti-diagonal test)**.** *One has*

$$\delta_{\mathrm{pair}} = \sigma_{\min}\Big(D_+|_{\mathcal{F}_{A,M}}\Big)^2, \qquad 0 \le \delta_{\mathrm{pair}} \le 2, \tag{66}$$

and

$$\begin{aligned} \delta_{\mathrm{pair}} > 0 &\iff \mathcal{F}_{A,M} \cap \{(g,-g) : g \in \mathcal{H}\} = \{(0,0)\} \\ &\iff \operatorname{Ker}(A+M) \subseteq \operatorname{Ker} A. \end{aligned} \tag{67}$$

Furthermore

$$\boxed{\delta(A,M) \ge \delta_{\mathrm{pair}}.} \tag{68}$$

Proof. The first identity is the Rayleigh quotient of $D_+^* D_+$ on the actual pair image. Its kernel is the anti-diagonal. If $Jz = (g,-g) \neq (0,0)$, then $Bz = 0$ and $Az = g \neq 0$; the converse is immediate. Finally, for $Az \neq 0$,

$$\frac{\|Bz\|^2}{\|Az\|^2} \ge \frac{\|Bz\|^2}{\|Az\|^2 + \|Mz\|^2} \ge \delta_{\mathrm{pair}}.$$

Taking the infimum proves (68). □

The pair formulation is often convenient for an exact matrix computation, but it must use $\operatorname{Ran} J$, not an independently chosen product of the represented and missing ranges. Enlarging the feasible set to $\operatorname{Ran} A \times \operatorname{Ran} M$ can manufacture an anti-diagonal pair which no physical coefficient vector produces.

4.2 Qualitative and quantitative separation

Example 4.4 (The kernel test has no quantitative content)**.** Let $A = I$ and $M = -(1-\varepsilon)I$, where $0 < \varepsilon < 1$. Then $B = \varepsilon I$, so

$$\operatorname{Ker} B = \operatorname{Ker} A = \{0\}, \qquad \delta(A,M) = \varepsilon^2. \tag{69}$$

The exact kernel test passes for every $\varepsilon > 0$, but the constant can be arbitrarily small. For an X-dependent family, injectivity at each finite X does not imply a polynomial lower bound.

Example 4.5 (Equal raw size, opposite reassembly)**.** Let $A = I$ and $M = \pm tI$, with $t \ge 0$. In both cases

$$\|Mz\|^2 = t^2 \|Az\|^2, \tag{70}$$

but

$$\delta(A, tI) = (1+t)^2, \qquad \delta(A, -tI) = (1-t)^2. \tag{71}$$

Thus the missing norm alone cannot distinguish constructive from destructive interference.

Example 4.6 (Pointwise success does not imply a uniform gate)**.** Return to Example 3.5. The distinguished vector $z_* = (1,0)$ satisfies

$$\frac{\|Bz_*\|^2}{\|Az_*\|^2} = 1, \tag{72}$$

while the uniform constant is zero because $(1,-1)$ cancels exactly. Any distinguished-vector claim must therefore remain explicitly labeled as such.

4.3 Scale-dependent interpretation

For a family (A_X, M_X), call $\lambda_{\text{reasm}} \geq 0$ an admissible uniform loss exponent if

$$\delta(A_X, M_X) \geq X^{-\lambda_{\text{reasm}}+o(1)}. \tag{73}$$

If the exact kernel test fails along an unbounded sequence of scales, no such finite exponent exists on that coefficient class. If the test passes, (73) is still a separate quantitative assertion. For one distinguished physical vector $\zeta_*(X)$, the analogous ratio

$$\delta_*(X) := \frac{\|B_X \zeta_*(X)\|^2}{\|A_X \zeta_*(X)\|^2} \tag{74}$$

is a different object and may remain positive after the uniform gate is killed.

5 Signed defects and two routes to reassembly

The Gram formulation is optimal for a coefficient-uniform theorem. For a distinguished physical coefficient vector, the exact remaining task is often clearer as a signed cross-term estimate.

5.1 The pointwise signed defect

Define

$$\mathfrak{X}(z) := 2\,\mathrm{Re}\langle Az, Mz\rangle + \|Mz\|^2. \tag{75}$$

Then

$$\boxed{\|Bz\|^2 = \|Az\|^2 + \mathfrak{X}(z).} \tag{76}$$

For $Az \neq 0$, this becomes

$$\boxed{\frac{\|Bz\|^2}{\|Az\|^2} = 1 + \frac{\mathfrak{X}(z)}{\|Az\|^2}.} \tag{77}$$

Consequently a desired pointwise transfer

$$\|Bz\|^2 \geq \delta_* \|Az\|^2 \tag{78}$$

is exactly equivalent to

$$\boxed{\mathfrak{X}(z) \geq -(1-\delta_*)\|Az\|^2.} \tag{79}$$

This is neither a positivity statement about $\|Mz\|^2$ nor an unsigned estimate for the atomic missing mass. The cross term is part of the physical problem.

In the prime-resolved notation of Proposition 2.3, $\mathfrak{X}$ is exactly the right side of (34). Hence its negative part can arise only from base–missing and unequal-prime native aliasing. The same-prime orthogonality is a genuine simplification, but it leaves the two surviving sums uncontrolled.

5.2 The nuisance-optimized defect

For $s \in S$, define the best defect obtainable by moving inside its A-fiber:

$$\mathfrak{X}_{\text{sh}}(s) := \inf_{k \in N} \left(\|B(s+k)\|^2 - \|A_S s\|^2 \right). \tag{80}$$

Proposition 5.1 (Shorted defect operator)**.** *With Q and R_S as in Section 3,*

$$\boxed{\mathfrak{X}_{\mathrm{sh}}(s) = \langle (Q - R_S)s, s\rangle.} \tag{81}$$

If

$$\Theta_{\mathrm{reasm}} := R_S^{-1/2}(Q - R_S)R_S^{-1/2}, \tag{82}$$

then

$$\boxed{\delta(A, M) = 1 + \lambda_{\min}(\Theta_{\mathrm{reasm}}).} \tag{83}$$

Proof. The least-squares identity (43) gives

$$\inf_{k\in N} \|B(s + k)\|^2 = \langle Qs, s\rangle.$$

Subtracting $\|A_S s\|^2 = \langle R_S s, s\rangle$ proves (81). Conjugating by $R_S^{-1/2}$ and using Theorem 3.3 proves (83). $\square$

Thus a coefficient-uniform signed route must control the negative spectrum of the *shorted* defect. Bounding the unshorted cross term only on the canonical sector S is insufficient when $B(N) \neq 0$.

5.3 An angle certificate

For $Az \neq 0$, put

$$t(z) := \frac{\|Mz\|}{\|Az\|}, \tag{84}$$

and, when $Mz \neq 0$, define the destructive cosine

$$\rho(z) := -\frac{\mathrm{Re}\langle Az, Mz\rangle}{\|Az\|\,\|Mz\|}. \tag{85}$$

Set $\rho(z) = 0$ if $Mz = 0$. Then $-1 \le \rho(z) \le 1$, and

$$\frac{\|Bz\|^2}{\|Az\|^2} = 1 + t(z)^2 - 2\rho(z)t(z). \tag{86}$$

Proposition 5.2 (Uniform angle certificate)**.** *Suppose there is $0 \le \rho_0 < 1$ such that*

$$\rho(z) \le \rho_0 \qquad \textit{whenever } Az \neq 0. \tag{87}$$

Then

$$\boxed{\delta(A, M) \ge 1 - \rho_0^2 > 0.} \tag{88}$$

In particular, if $\mathrm{Re}\langle Az, Mz\rangle \ge 0$ for all z, then $\delta(A, M) \ge 1$.

Proof. If $\rho(z) \ge 0$, complete the square:

$$1 + t^2 - 2\rho t = (t - \rho)^2 + 1 - \rho^2 \ge 1 - \rho_0^2.$$

If $\rho(z) < 0$, the left side is at least one. Take the infimum over z. $\square$

This angle route may succeed even when the missing norm is not small. It is also substantially more arithmetic: one must retain the phases in the base–missing and unequal-prime alias sums. The mass–degree route in the next section is cruder, but it replaces phase control by a quantitative smallness condition.

6 Raw missing mass times native fiber degree

We now give a phase-blind sufficient condition. It is sharp at the level of information it uses.

6.1 The exact grouping norm

Let $\mathcal{W}_M$ be a finite missing atomic carrier, let $\mathcal{I}_{\mathrm{out}}$ be the active native output set, and let

$$\mathcal{F}_i^{\mathrm{miss}} := \{w \in \mathcal{W}_M : i(w) = i\}. \tag{89}$$

Absorb every fixed unit Möbius or high-character phase into the atomic amplitude $m_w(z)$, and define

$$L_M : \mathcal{K} \longrightarrow \ell^2(\mathcal{W}_M), \qquad (L_M z)_w := m_w(z), \tag{90}$$

$$C_M : \ell^2(\mathcal{W}_M) \longrightarrow \ell^2(\mathcal{I}_{\mathrm{out}}), \qquad (C_M a)_i := \sum_{w \in \mathcal{F}_i^{\mathrm{miss}}} a_w. \tag{91}$$

Then

$$M = C_M L_M, \qquad \mathcal{D}_{\mathrm{miss}}(z) := \|L_M z\|_2^2. \tag{92}$$

In the fixed-shift specialization, every absorbed phase has modulus one, so $\mathcal{D}_{\mathrm{miss}}(z) = \mathcal{D}_M(z)$ with the raw diagonal of (16). Put

$$d_{\mathrm{miss}} := \max_{i \in \mathcal{I}_{\mathrm{out}}} |\mathcal{F}_i^{\mathrm{miss}}|, \tag{93}$$

with $d_{\mathrm{miss}} = 0$ when the missing carrier is empty.

Proposition 6.1 (Exact native grouping norm)**.** *One has*

$$\boxed{\|C_M\|^2 = d_{\mathrm{miss}},} \tag{94}$$

and therefore, for every $z \in \mathcal{K}$,

$$\boxed{\|Mz\|^2 \le d_{\mathrm{miss}} \mathcal{D}_{\mathrm{miss}}(z).} \tag{95}$$

Proof. Fiberwise Cauchy–Schwarz gives

$$|(C_M a)_i|^2 \le |\mathcal{F}_i^{\mathrm{miss}}| \sum_{w \in \mathcal{F}_i^{\mathrm{miss}}} |a_w|^2.$$

Summing over the disjoint output fibers proves $\|C_M\|^2 \le d_{\mathrm{miss}}$. On a fiber of maximal size, choose all coordinates equal with a common phase and set all other coordinates to zero. Equality follows. Applying the operator norm bound to $L_M z$ proves (95). □

The exact norm concerns the ambient grouping map. The physical range $\operatorname{Ran} L_M$ may have a smaller restricted norm, but proving that is another Gram problem. Conversely, replacing d_{miss} by an average degree is invalid for a coefficient-uniform estimate.

6.2 Distinguished-vector certificate

Theorem 6.2 (Sharp mass–degree lower certificate)**.** *For every z with $Az \neq 0$,*

$$\boxed{\frac{\|(A+M)z\|^2}{\|Az\|^2} \ge \left(1 - \sqrt{\frac{d_{\mathrm{miss}} \mathcal{D}_{\mathrm{miss}}(z)}{\|Az\|^2}}\right)_+^2.} \tag{96}$$

In particular, if

$$d_{\mathrm{miss}} \mathcal{D}_{\mathrm{miss}}(z) \le \eta \|Az\|^2, \qquad 0 \le \eta < 1, \tag{97}$$

then the ratio is at least $(1 - \sqrt{\eta})^2$.

Proof. The reverse triangle inequality and Proposition 6.1 give

$$\|(A+M)z\| \geq (\|Az\| - \|Mz\|)_+ \geq \left(\|Az\| - \sqrt{d_{\mathrm{miss}}\mathcal{D}_{\mathrm{miss}}(z)}\right)_+ .$$

Divide by $\|Az\|$ and square. $\square$

Example 6.3 (Sharp cancellation at the product threshold)**.** Take one represented native output $Az = 1$. Let one native fiber contain n missing atoms, each of amplitude $-1/n$. Then

$$d_{\mathrm{miss}} = n, \qquad \mathcal{D}_{\mathrm{miss}}(z) = \frac{1}{n}, \qquad Mz = -1. \tag{98}$$

Thus $d_{\mathrm{miss}}\mathcal{D}_{\mathrm{miss}}/\|Az\|^2 = 1$ and the full output vanishes. The same construction with $n = 2$ shows that very small combinatorial degree is already compatible with exact cancellation. Therefore neither $\mathcal{D}_{\mathrm{miss}} = o(1)$ nor a bounded degree alone is a sufficient reassembly theorem.

6.3 Uniform and canonical-sector versions

Corollary 6.4 (Uniform mass–degree certificate)**.** *Suppose $\varepsilon \geq 0$ and*

$$\mathcal{D}_{\mathrm{miss}}(z) \leq \varepsilon\|Az\|^2 \qquad \textit{for every } z \in \mathcal{K}. \tag{99}$$

Then $M|_{\operatorname{Ker} A} = 0$ and

$$\boxed{\delta(A, M) \geq (1 - \sqrt{d_{\mathrm{miss}}\varepsilon})_+^2.} \tag{100}$$

Proof. If $z \in \operatorname{Ker} A$, (99) forces $\|L_M z\| = 0$, hence $Mz = 0$. Apply Theorem 6.2 to every z with $Az \neq 0$ and take the infimum. $\square$

The hypothesis in Corollary 6.4 is strong: it automatically removes the nuisance image. When only the canonical sector can be controlled, the following version retains the missing transversality explicitly. With the notation of Section 3, define

$$\gamma_A := \inf_{s \in S\setminus\{0\}} \frac{\|PA_S s\|^2}{\|A_S s\|^2}. \tag{101}$$

Proposition 6.5 (Canonical mass plus nuisance transversality)**.** *Suppose $\varepsilon \geq 0$ and*

$$\mathcal{D}_{\mathrm{miss}}(s) \leq \varepsilon\|A_S s\|^2 \qquad (s \in S). \tag{102}$$

Then

$$\boxed{\delta(A, M) \geq \left(\sqrt{\gamma_A} - \sqrt{d_{\mathrm{miss}}\varepsilon}\right)_+^2 .} \tag{103}$$

Proof. For $s \in S$,

$$\begin{aligned}
\|PB_S s\| &= \|P(A_S s + Ms)\| \\
&\geq \|PA_S s\| - \|PMs\| \\
&\geq \left(\sqrt{\gamma_A} - \sqrt{d_{\mathrm{miss}}\varepsilon}\right)\|A_S s\|.
\end{aligned}$$

Use Theorem 3.1 and take the positive part. $\square$

This estimate exposes two independent risks: the missing atomic output can be large, and the represented output can already lie close to the nuisance range $B(\operatorname{Ker} A)$. An estimate for only one of them does not close the uniform gate.

6.4 Exponent form

For a distinguished vector $z_*(X)$, write

$$\frac{\|B_X z_*(X)\|^2}{\|A_X z_*(X)\|^2} \geq X^{-\lambda_{\mathrm{reasm},*}+o(1)} \tag{104}$$

when the denominator is nonzero. The star distinguishes this pointwise exponent from the coefficient-uniform λ_{reasm} of Section 4. Suppose now, for that distinguished vector or uniformly on a stated class,

$$d_{\mathrm{miss}} = X^{\mu_{\mathrm{miss}}+o(1)}, \qquad \frac{\mathcal{D}_{\mathrm{miss}}(z)}{\|Az\|^2} \leq X^{-\sigma_{\mathrm{miss}}+o(1)}. \tag{105}$$

Writing

$$q_X(z) := \frac{d_{\mathrm{miss}}\mathcal{D}_{\mathrm{miss}}(z)}{\|Az\|^2}, \tag{106}$$

the triangle inequality gives the two-sided estimate

$$(1-\sqrt{q_X(z)})_+^2 \leq \frac{\|Bz\|^2}{\|Az\|^2} \leq (1+\sqrt{q_X(z)})^2. \tag{107}$$

Consequently

$$\sigma_{\mathrm{miss}} > \mu_{\mathrm{miss}} \quad \Longrightarrow \quad \frac{\|Bz\|^2}{\|Az\|^2} = 1 + o(1) \tag{108}$$

Thus the mass–degree route gives $\lambda_{\mathrm{reasm},*} = 0$ in the distinguished-vector scope and $\lambda_{\mathrm{reasm}} = 0$ when the hypotheses hold uniformly on the declared class. At equality of the exponents, the exact sufficient condition furnished by this certificate is

$$(1-\sqrt{q_X(z)})_+^2 \geq X^{-o(1)}. \tag{109}$$

This bound must hold pointwise for the distinguished vector, or uniformly over the declared coefficient class, according to the claimed scope. The simpler condition $\limsup q_X(z) < 1$ is sufficient but not necessary; for example, $q_X = 1 - 1/\log X$ still gives a subpower lower bound. Thus exponent equality requires finer constant or subpower control. If $\sigma_{\mathrm{miss}} < \mu_{\mathrm{miss}}$, the certificate is silent; it does not prove failure.

More generally, if a common reference diagonal satisfies

$$\mathcal{D}_{\mathrm{miss}}(z) \leq X^{-\sigma+o(1)}\mathcal{D}_{\mathrm{ref}}(z), \qquad \|Az\|^2 \geq X^{-\lambda_{\mathrm{rep}}+o(1)}\mathcal{D}_{\mathrm{ref}}(z), \tag{110}$$

then the zero-cost sufficient condition becomes

$$\boxed{\sigma > \mu_{\mathrm{miss}} + \lambda_{\mathrm{rep}}.} \tag{111}$$

Every exponent in this comparison refers to the same physical vector and the same raw reference measure.

7 Arithmetic localization of the missing carrier

The previous sections are exact for any pair of finite-dimensional maps. We now return to the fixed-h_0 Walsh carrier and record what the literal arithmetic dictionary does and does not add.

7.1 Boolean signatures and ordered first failure

For $w \in \mathcal{W}_H(y)$, recover m, j, γ, n, p, r, s by (19)–(20). Membership in the represented terminal image is a finite conjunction of declared tests. For definiteness, use the dependency order

$$T_1(w) := \mathbf{1}_{\{(m,r)\in\mathcal{I}_X\}}, \tag{112}$$
$$T_2(w) := \mathbf{1}_{\{j,\gamma,p \text{ lie in the declared terminal scopes}\}}, \tag{113}$$
$$T_3(w) := \mathbf{1}_{\{\Lambda_X(n)\neq 0\}}, \tag{114}$$
$$T_4(w) := \mathbf{1}_{\{d_m,r,s \text{ squarefree and } (d_m,r)=1\}}, \tag{115}$$
$$T_5(w) := \mathbf{1}_{\{\text{all remaining terminal-only masks pass}\}}. \tag{116}$$

The lists inside T_2 and T_5 are the inherited literal declarations, not new favorable supports. By construction,

$$w \in \mathcal{W}_R \iff T_1(w) = \cdots = T_5(w) = 1. \tag{117}$$

The Boolean signature $(T_1, \ldots, T_5)$ is order independent. A "first failure" is not: it becomes well defined only after the order above is declared. Define

$$\mathcal{F}_1 := \{w \in \mathcal{W}_H : T_1(w) = 0\}, \tag{118}$$
$$\mathcal{F}_\nu := \{w \in \mathcal{W}_H : T_1(w) = \cdots = T_{\nu-1}(w) = 1, T_\nu(w) = 0\}, \quad 2 \le \nu \le 5. \tag{119}$$

Proposition 7.1 (Exact ordered missing partition)**.** *The first-failure sets are disjoint and*

$$\boxed{\mathcal{W}_M = \bigsqcup_{\nu=1}^{5} \mathcal{F}_\nu.} \tag{120}$$

If

$$\mathcal{D}_\nu(\zeta) := \sum_{w\in\mathcal{F}_\nu} |b_w(\zeta)|^2, \tag{121}$$

then

$$\boxed{\mathcal{D}_M = \sum_{\nu=1}^{5} \mathcal{D}_\nu.} \tag{122}$$

Proof. Every missing atom fails at least one test by (117); its least failing index is unique. Conversely every atom in a first-failure set fails terminal admission and is missing. Raw atomic additivity gives the energy identity. □

The adjective "ordered" is essential. Reordering the tests changes the first-failure labels, although it does not change $\mathcal{W}_M$ or the full Boolean signature.

7.2 Empty pseudo-failure layers

Several apparent failure mechanisms are already excluded on the common active Walsh carrier.

Proposition 7.2 (Inherited automatic tests)**.** *For an actual nonzero high Walsh atom on the common TPC–45/TPC–53 physical carrier:*

- *a* $p = P^+(n) > y$*,* $pr = mj + h_0$*, and* $r = n/p$ *hold by the unique high-prime decoding;*
- *b the nonzero Walsh amplitude already enforces* $\Lambda_X(n) \neq 0$ *and its shared literal profile masks;*
- *c the inherited squarefree and coprimality support implies that* r *and* $s = d_m r$ *are squarefree and* $(d_m, r) = 1$*.*

Consequently the first-failure layers created solely by T_3 and T_4 are empty. The part of T_5 consisting only of masks shared with the Walsh carrier is empty as well.

Proof. Part a is the largest-prime census and the assumption $y > \sqrt{N_+}$. For b, coordinates with zero shaped weight or zero shared profile were removed as identically inactive when the common Walsh carrier was formed. For c, the TPC–45 support makes $d_m n$ squarefree and $(d_m, n) = 1$. Since $n = pr$, every divisor r is squarefree, is coprime to d_m, and therefore $s = d_m r$ is squarefree. □

Corollary 7.3 (Identical-support specialization)**.** *Suppose the Walsh and terminal constructions use identical row, orbit, opposite-row, terminal-prime, literal-mask, and profile supports, and there is no additional terminal-only pruning. Then*

$$\boxed{\mathcal{W}_M = \{w \in \mathcal{W}_H(y) : (m(w), r(w)) \notin \mathcal{I}_X\}.} \tag{123}$$

Thus the only missing-high mechanism in this specialization is failure of the complementary cofactor to enter the retained-pair carrier.

This is a support-alignment theorem, not an occupancy theorem. It does not show that the set in (123) has small mass, or even that some active row contains a represented atom.

7.3 Zero-cost localization of the diagnostic branch

Let $q \leq 5$ be the number of nonempty first-failure layers, and write

$$N_{\text{dyad}}(y) := 1 + \left\lfloor \log_2 \frac{N_+}{y} \right\rfloor = O(\log X) \tag{124}$$

for the number of represented cofactor cells. If $\mathcal{D}_M > 0$, one of them satisfies

$$\max_{\nu} \mathcal{D}_\nu \geq \frac{\mathcal{D}_M}{q}. \tag{125}$$

Combining this constant partition with the $O(\log X)$ dyadic partition of the represented terminal face gives the exact dichotomy

$$\mathcal{D}_H^W := \mathcal{D}_R + \mathcal{D}_M > 0 \quad \Longrightarrow \quad \begin{cases} \exists k : \mathcal{D}_k \geq \dfrac{\mathcal{D}_H^W}{2N_{\text{dyad}}(y)}, & \mathcal{D}_R \geq \mathcal{D}_H^W/2, \\ \exists \nu : \mathcal{D}_\nu \geq \dfrac{\mathcal{D}_H^W}{2q}, & \mathcal{D}_M \geq \mathcal{D}_H^W/2, \end{cases} \tag{126}$$

Hence localization has zero fixed-power cost in either branch:

$$\boxed{\lambda_{\text{part}} = 0.} \tag{127}$$

The conclusion is diagnostic. It either finds a large represented block or identifies a dominant reason for missing mass. It does not force the represented alternative and does not imply $\lambda_{\text{reasm}} = 0$.

7.4 Exact rowwise represented-mass minimax

Write

$$W_E(m) := \sum_{\substack{w \in \mathcal{W}_E \\ m(w) = m}} |\beta_w|^2, \qquad E \in \{R, M, H\}, \qquad W_H := W_R + W_M. \tag{128}$$

Then literal factorization gives

$$\mathcal{D}_E(\zeta) = \sum_m W_E(m) |\zeta_m|^2. \tag{129}$$

Proposition 7.4 (Sharp represented-mass minimax)**.** *The optimal constant uniform over any bounded coefficient cube containing a nonzero multiple of each coordinate direction is*

$$\boxed{\inf_{\mathcal{D}_H^W(\zeta)>0} \frac{\mathcal{D}_R(\zeta)}{\mathcal{D}_H^W(\zeta)} = \min_{m:W_H(m)>0} \frac{W_R(m)}{W_H(m)}.} \tag{130}$$

In particular, if one active row has $W_R(m) = 0 < W_H(m)$, the uniform represented-mass transfer is exactly zero.

Proof. The quotient is a weighted average of the row ratios $W_R(m)/W_H(m)$, with nonnegative weights $W_H(m)|\zeta_m|^2$. It is at least their minimum. Concentrating ζ on a minimizing row attains that minimum after a harmless scalar rescaling into the bounded cube. □

Thus a geometric density statement about the union of rows cannot replace a rowwise lower bound when the coefficient class has only upper bounds on $|\zeta_m|$.

7.5 Residual ladders and ambient degree

Fix a common source-prime cell of diameter $O(L)$, a residual key (i, s), and a dyadic cofactor block $R \le r < 2R$. Write

$$m = d\ell, \qquad r = \frac{s}{d}. \tag{131}$$

The fixed-shift equality becomes

$$rp - dj\ell = h_0. \tag{132}$$

Proposition 7.5 (Ambient residual ladder bound)**.** *On one common D, L cell, the number of full, represented, missing, or first-failure high atoms with fixed residual key (i, s) and $R \le r < 2R$ satisfies*

$$\boxed{d^{\mathrm{res}}_{\max}(R) \ll_{h_0} \tau(s)\left(1 + \frac{L}{R}\right) = X^{o(1)}\left(1 + \frac{L}{R}\right).} \tag{133}$$

Moreover the prime-resolved degrees at (i, p) and (i, s, p) are at most one. The direct native degree has only the inherited crude bound

$$d_{\mathrm{miss}} \ll X^{o(1)}Q. \tag{134}$$

Proof. Fix $d \mid s$, hence $r = s/d$, and put $g = (r, dj) = (r, j)$, using $(d, r) = 1$. If (132) has one integral solution (p_0, ℓ_0), all integral solutions lie on the determinant ladder

$$p = p_0 + \frac{dj}{g}t, \qquad \ell = \ell_0 + \frac{r}{g}t, \qquad t \in \mathbb{Z}. \tag{135}$$

Solvability forces $g \mid h_0$, and therefore $g \le |h_0|$. Since ℓ lies in an interval of diameter $O(L)$, the ladder contributes $O_{h_0}(1 + L/R)$ candidates. Summing over $d \mid s$ gives (133); primality, retained-pair, residue, mask, and profile tests only delete candidates. The fixed- (i, s, p) assertion is the TPC–55/TPC–58 prime-resolved injection, while the stronger fixed-(i, p) assertion is the TPC–59 critical-cutoff row singleton proved again in Proposition 2.3.

For the native bound, fixing i fixes the side, opposite row, and j. Fixed-coordinate fused-label injectivity leaves at most one active high atom per source row, and the inherited source-system support bound is $\#\mathcal{A}_\sigma \ll X^{o(1)}Q$, $\sigma \in \{1, 2\}$. This proves (134). A wider union of D, L cells must pay explicitly for the additional cells. □

The residual estimate is not the native estimate used directly in (95): erasing s can merge many residual fibers. Prime-resolved singleton structure therefore does not imply native singleton structure.

8 Endpoint audit and the reassembly ledger

The exact constants above identify the reassembly problem. We now compare the available coefficient-blind degree certificate with the endpoint budget and state the admissible ways to close the new gate.

8.1 Critical-cutoff exponent dictionary

Assume that y is very-high admissible and write

$$\frac{y}{Q} = X^{\xi+o(1)}, \qquad 0 \le \xi < \frac{133}{400}. \tag{136}$$

At $\xi = 0$, admissibility retains the constant-scale requirement $y \ge C_{\mathrm{vh}} Q$ for a sufficiently large support-dependent constant; the exponent identity alone is not enough. The displayed range is the one in which a polynomially nonempty cofactor scale remains compatible. The compatible cofactor ceiling is

$$r \le \frac{N_+}{y} = X^{\theta_c(\xi)+o(1)}, \qquad \theta_c(\xi) := \frac{133}{400} - \xi. \tag{137}$$

The inherited source-prime and divisor exponents are

$$\lambda_L := \frac{267}{400} - \frac{10049}{52500} = \frac{99979}{210000}, \qquad \lambda_D := \frac{10049}{52500}. \tag{138}$$

Thus the residual label $s = d_m r$ obeys

$$s \le X^{\lambda_D + \theta_c(\xi) + o(1)} = X^{110021/210000 - \xi + o(1)}. \tag{139}$$

Let $R = X^{\theta_R + o(1)}$ be any compatible dyadic cofactor block, so

$$0 \le \theta_R \le \theta_c(\xi). \tag{140}$$

The determinant-ladder estimate (133) supplies the coefficient-blind capacity exponent

$$\mu_{\mathrm{cap}}(R) := \lambda_L - \theta_R. \tag{141}$$

It satisfies

$$\boxed{\mu_{\mathrm{cap}}(R) \ge \frac{15077}{105000} + \xi,} \tag{142}$$

and hence

$$\boxed{\mu_{\mathrm{cap}}(R) - \frac{1}{400} \ge \frac{29629}{210000} + \xi.} \tag{143}$$

At the critical choice $\xi = 0$, the lower floor is approximately 0.143590, whereas the entire endpoint allowance is 0.0025.

This calculation must be interpreted in the correct direction. μ_{cap} is the exponent of an ambient *upper* bound for a fiber size; it is not a proved energy loss and does not say that actual fibers attain that size. It does show that a crude route which pays the complete ambient degree through coloring, a coefficient-blind transversal, or Cauchy–Schwarz cannot fit inside the 1/400 ledger. That support-only closure is therefore unavailable. A successful argument must use actual missing-mass sparsity, restricted-range geometry, or signed phase information.

8.2 Three admissible reassembly certificates

The following routes preserve the physical coefficient vector and the fixed shift.

1. *Shorted-Gram route.* Prove directly that

$$\lambda_{\min}\left(R_S^{-1/2} Q R_S^{-1/2}\right) \geq X^{-\lambda_{\mathrm{reasm}}+o(1)}. \tag{144}$$

This is the optimal coefficient-uniform statement. Before estimating an eigenvalue, apply the exact kernel and dimension kill tests.

2. *Signed distinguished-vector route.* For the physical coefficient vector ζ_*, prove

$$\mathfrak{X}(\zeta_*) \geq -\left(1 - X^{-\lambda_{\mathrm{reasm},*}+o(1)}\right) \|A_X \zeta_*\|^2. \tag{145}$$

This may remain open after the uniform gate is killed, but it cannot be promoted to a uniform claim.

3. *Mass–degree route.* Prove a common-reference saving σ, a represented loss λ_{rep}, and a native degree exponent μ_{miss} satisfying

$$\sigma > \mu_{\mathrm{miss}} + \lambda_{\mathrm{rep}}. \tag{146}$$

Then Theorem 6.2 yields $\lambda_{\mathrm{reasm}} = 0$ for a uniform scope, or $\lambda_{\mathrm{reasm},*} = 0$ for the distinguished-vector scope. At equality of exponents, finer constant or subpower control must verify (109); the exponent comparison alone is inconclusive.

The angle certificate of Proposition 5.2 is a fourth presentation of the signed route. Orthogonal full represented-affine and missing output ranges, $\operatorname{Ran} A \perp \operatorname{Ran} M$, give the safe special case $\delta(A, M) \geq 1$, but the fixed-h_0 maps here do not currently satisfy such an orthogonality theorem.

8.3 Complete ledger

The prospective fixed-shift chain remains

$$\begin{aligned} &\mathcal{D}_{\mathrm{up}} \longrightarrow \mathcal{E}_{\mathrm{par}} \longrightarrow \mathcal{E}_{\mathcal{R}} \longrightarrow \mathcal{D}^{\mathrm{term}}_{\mathcal{R}} \longrightarrow \mathcal{D}^{\mathrm{rep}}_{H} \longrightarrow \mathcal{D}_{k_*} \\ &\quad \longrightarrow \mathcal{G}_H \longrightarrow \mathcal{N}_H \longrightarrow \mathcal{N}^{\mathrm{rep}}_{\mathrm{aff}} \xrightarrow{\text{reassembly}} \mathcal{N}^{\mathrm{full}}_{\mathrm{slice}} \xrightarrow{\text{boundary/tail}} \text{parent arithmetic target.} \end{aligned} \tag{147}$$

Every arrow is a different assertion. TPC–60 replaces the displayed reassembly arrow by the exact constant $\delta(A_X, M_X)$; it does not estimate that constant from below for the arithmetic family.

If all arrows eventually receive polynomial lower bounds, the energy loss is

$$\boxed{\begin{aligned} \Lambda_{\mathrm{tot}} := \lambda_{\mathrm{sel}} + \lambda_{\mathrm{occ}} + \lambda_{\mathrm{blk}} + \lambda_{\mathrm{term}} + \lambda_{\mathrm{sing}} \\ + \lambda_H + \lambda_{\mathrm{aff}} + \lambda_{\mathrm{reasm}} + \lambda_{\mathrm{bdry}} + \lambda_{\mathrm{tail}}. \end{aligned}} \tag{148}$$

The endpoint requirement is the strict inequality

$$\boxed{\Lambda_{\mathrm{tot}} < \frac{1}{400}.} \tag{149}$$

For a distinguished-vector chain, λ_{reasm} in (148) is replaced consistently by $\lambda_{\mathrm{reasm},*}$. The two exponents are never interchanged: a pointwise certificate cannot be entered as a uniform one. Equality is rejected unless leading constants and all remainders are quantified. The cutoff $y \asymp Q$ is a feasibility choice and is not an additive energy loss. TPC–59 proves $\lambda_{\mathrm{blk}} = 0$ only after represented high-face activation and with its distinguished-block scope. The new identity $\lambda_{\mathrm{part}} = 0$ is a diagnostic finite/logarithmic partition and does not replace either reassembly exponent in its matching version of (148).

8.4 Stop rules

The current route must stop or change scope under any of the following events.

1. A literal vector satisfies $A_X\zeta \neq 0$ and $B_X\zeta = 0$; then the uniform reassembly gate is exactly dead.

2. An active row satisfies $W_R(m) = 0 < W_H(m)$; then uniform represented raw-mass transfer is exactly zero.

3. A necessary proved loss by itself reaches $1/400$, or the sum of necessary losses reaches it without a constant-level endpoint argument.

4. The only available proof pays the full ambient capacity exponent in (142); then that proof architecture exceeds the endpoint allowance even at the critical cutoff.

None of these events may be repaired by changing h_0, averaging shifts, renormalizing atoms by their fiber sizes, deleting coefficient-dependent zeros, or selecting represented and missing vectors independently. Such operations define different problems.

9 Claims, limitations, and the next arithmetic gate

9.1 Result ledger

Object	Exact conclusion	Status
Three physical maps	One row-coefficient space and one fixed h_0 support $B_X = A_X + M_X$, with raw atomic $\mathcal{D}_0 = \mathcal{D}_B + \mathcal{D}_R + \mathcal{D}_M$.	L1 identity
Prime-resolved alias	Same-prime represented and missing vectors have disjoint support; only base–missing interaction and unequal-prime aliasing remain.	L1 identity
Uniform reassembly quotient	Exact nuisance elimination, shorted-Gram generalized eigenvalue, and effective least singular value.	L0 theorem
Uniform zero test	$\delta(A, M) > 0$ exactly when $\operatorname{Ker}(A + M) \subseteq \operatorname{Ker} A$; a dimension excess kills it.	L0 theorem
Distinguished-vector defect	Exact signed identity $\|Bz\|^2 - \|Az\|^2 = \mathfrak{X}(z)$.	L0 identity
Raw mass–degree bridge	$\|Mz\|^2 \leq d_{\mathrm{miss}}\mathcal{D}_{\mathrm{miss}}(z)$, with a sharp reverse-triangle certificate.	L0 theorem
Missing reason partition	Definitional ordered first-failure partition; the inherited-support theorem makes the automatic pseudo-failure layers empty on the common literal carrier.	L1 definitional partition plus inherited-support theorem
Identical-support specialization	Missing high atoms are exactly those whose complementary pair (m, r) is outside $\mathcal{I}_X$.	Conditional L1 identity
Localization	One represented dyadic block or one missing failure layer retains an $X^{-o(1)}$ fraction of its branch.	L1, $\lambda_{\mathrm{part}} = 0$
Endpoint certificate	A route that pays the full residual determinant-ladder capacity incurs exponent at least $15077/105000 + \xi$, far above $1/400$; this is not a lower bound for the actual native degree.	Route-specific no-go

9.2 What is not proved

The following statements do not follow from this paper:

1. $\mathcal{D}_M = o(\mathcal{D}_R)$, either rowwise or for the distinguished coefficient vector;
2. a polynomial lower bound for $\delta(A_X, M_X)$;
3. a favorable sign for the base–missing or unequal-prime cross terms;
4. $\lambda_{\text{reasm}} = 0$ for the physical TPC family;
5. a positive fixed-h_0 parity gain, a prime-pair lower bound, a prime-pair asymptotic, or a twin-prime consequence.

The first-failure partition can identify where missing mass lies, but does not make that mass small. The same-prime orthogonality deletes one class of cross terms, but different primes still alias at the native output. A finite-scale positive eigenvalue is numerical scouting evidence unless its scale dependence is controlled.

9.3 The next ordered attack

The next paper should not introduce another abstract angle. The exact reassembly theorem leaves a concrete arithmetic decision tree.

1. *Rowwise census.* Compute or bound the literal weights
$$W_R(m), \qquad W_M(m), \qquad \frac{W_R(m)}{W_R(m) + W_M(m)} \tag{150}$$
on every active row. If an active missing-only row occurs, record the exact death of the uniform represented-mass route.

2. *Cofactor-defect estimate.* Under the identical-support specialization, replace the vague terminal failure by the single event
$$(m, r) \notin \mathcal{I}_X. \tag{151}$$
Estimate its coefficient-weighted mass relative to the represented affine reference, not merely its unweighted cardinality.

3. *Restricted native degree.* Factor the missing map through residual keys and estimate its norm on the actual coefficient image. The ambient bound $X^{o(1)}Q$ is too large, while the residual ladder bound alone does not survive erasure of s. A successful argument must prove either a smaller restricted degree exponent or destructive phase cancellation inside the missing map itself.

4. *Shorted kernel audit.* Assemble the literal matrices A_X, M_X at finite scales and test
$$\operatorname{Ker}(A_X + M_X) \subseteq \operatorname{Ker} A_X, \qquad \dim\big((\operatorname{Ker} A_X)^{\perp}\big) \le \dim \mathcal{H}_{\text{out}} - \operatorname{rank}(B_X|_{\operatorname{Ker} A_X})\,. \tag{152}$$
Repeated exact failure closes the uniform door. Repeated numerical success does not prove an asymptotic lower singular value, but it can identify the correct subspace for a theorem.

5. *Signed fallback.* If the uniform door is killed, retain the physical distinguished vector and estimate
$$-2\,\mathrm{Re}\langle A_X\zeta_*, M_X\zeta_*\rangle \tag{153}$$
using the exact base–missing and unequal-prime decomposition. This is a legitimate narrower route, provided it is not relabeled uniform.

The mass–degree route advances automatically at zero fixed-power cost when its arithmetic saving strictly beats the degree plus all represented-reference losses, as in (111). At equality, the finer test (109) decides whether this certificate still closes at subpower cost. If every provable lower bound already spends the remaining $1/400$ slack, the correct result is a route-closure theorem, not an adjusted normalization.

Conclusion. TPC–59 exposed a high Walsh boundary omitted by the represented terminal census. The present paper closes the corresponding interface: it writes the physical maps on one coefficient space, identifies the precise same-prime simplification, eliminates invisible coefficient directions by shorting the full Gram, and gives sharp kernel and mass–degree certificates. The route has therefore moved from an unspecified *off-carrier restoration* to a finite list of falsifiable arithmetic tasks. It has not crossed the parity barrier.

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